Macroscopic Crowd Control with Direction Constraints and Entrance-Rate Control for Urban Pedestrian Zones

Author Name

***Abstract*—Open scenic areas and major event venues often face systematic stampede risks caused by highly concentrated pedestrian flows, while existing large-scale crowd control methods largely rely on empirical rules and post-event reviews, making active prevention and quantitative evaluation difficult. In such scenarios, the principal operational controls are to set channel directions, including one-way, two-way, and closed states, and to regulate time-varying entrance release intensities; however, the coupled effects of these two types of controls on global pedestrian flow still lack quantitative modeling and optimization tools. To address this problem, this paper proposes a macroscopic crowd dynamics modeling and control-measure optimization method for fine-grained large-scale crowd management. The model is built upon the Hughes continuum framework. To represent operational rules in a continuum setting, local admissible direction sets are introduced to characterize discrete control states such as one-way movement, two-way movement, and channel closure, while anisotropic mobility tensors describe the geometric guidance exerted by channel layouts on pedestrian flow. On this basis, internal channel entrance flow-rate control is further introduced, enabling a control strategy to jointly specify travel directions and time-varying release intensities. To solve the resulting mixed-variable optimization problem composed of discrete direction configurations and continuous release intensities, this paper designs a Hierarchical Constrained Mixed-variable Black-box Optimization (HCMBO) algorithm for searching optimal crowd control strategies. Experiments on a simplified scenic-platform scenario show that the proposed macroscopic model can capture the complex interactions among multi-stage behavior, channel geometric guidance, and direction constraints. Under a composite objective formed by weighted indicators including total travel time, high-density exposure, and realized channel-load imbalance, repeated experiments show that HCMBO achieves the lowest mean objective value. It reduces the objective by approximately 4.0% compared with the second-best TPE-style mixed Bayesian optimization method, and also outperforms random search, pure simulated annealing, differential evolution, and other baseline methods. The proposed method provides a quantitative and interpretable decision-support tool for crowd routing and entrance flow control in open public spaces.**

***Index Terms*—Macroscopic crowd dynamics, modeling, channel optimization, Hamilton-Jacobi-Bellman equation, bi-level optimization.**

I. Introduction

LARGE-SCALE crowd gathering risks not only threaten individual safety but may also escalate into stampede incidents that disrupt social order and public security; the effectiveness of crowd management directly tests urban governance capacity and the modernization level of social safety governance systems [1], [2]. With the increasing frequency of high-density pedestrian activities such as holiday tourism, urban public events, commercial district promotions, New Year celebrations, and major sports events, core urban areas must accommodate pedestrian demands far exceeding daily levels within short time windows, posing enormous challenges for on-site organization, passage management, risk early warning, and emergency dispatch [3], [4], [5]. Against this backdrop, scientific modeling, numerical simulation, and optimization methods for formulating and evaluating crowd control strategies—improving the efficiency of large-scale event organization and reducing local congestion risks—have become critical scientific problems demanding urgent breakthroughs in social safety and urban governance [6], [7].

Crowds in large-scale events constitute typical complex systems. Numerous individuals with heterogeneous desired speeds, destinations, and route preferences interact through multi-scale, nonlinear mechanisms in confined spaces, generating spatiotemporal patterns at the collective level [8], [4]. Macroscopically, these patterns manifest as destination-oriented flow, passage convergence, bottleneck queuing, exit competition, and directional responses to control regulations [9], [10]; microscopically, individuals are influenced by surrounding density, visible paths, obstacle boundaries, and guidance facilities [11]. For open scenic areas, crowd behavior additionally exhibits pronounced phase dependence: visitors may first move toward the viewing platform, then tour along the main viewing direction, and subsequently choose different exit passages based on preferences [12]. These characteristics of multi-stage, multi-route coexistence with shared congestion mean that traditional static capacity analysis or single shortest-path models cannot adequately support refined management decisions.

Among various urban crowd gathering scenarios, open scenic areas, waterfront viewing platforms, and historic commercial districts are highly representative. Unlike enclosed venues such as stadiums and subway stations with well-defined boundaries [13], [14], such areas consist of open spaces, main tourist corridors, multiple lateral connecting passages, and return flow lines; visitors typically undergo a continuous behavioral chain of “entry–touring along the main corridor–choosing passages to leave–returning” [15], with crowd movement governed jointly by spatial geometry, control regulations, and historical behavioral preferences [16], [15]. Taking the Shanghai Nanjing East Road–Bund area as a concrete example, visitors approach the Bund from Nanjing East Road, ascend to the viewing platform via multiple stepped passages, tour along the waterfront, then exit through several southern passages and return along lower streets. This scenario can be abstracted as the typical “waterfront platform–multi-stepped passage” crowd flow scenario. In such scenarios, management

The author is with the Department, Institution, City, Country (e-mail: email@address).

(a) (b)

Fig. 1. Field-control context of the Shanghai Nanjing East Road–Bund area. (a) Large-scale holiday crowd under one-way passage organization. (b) Entrance-capacity control at a main visitor inflow location.

difficulties manifest in three aspects: first, multiple passages simultaneously serve entry, exit, and diversion functions, and local direction settings can alter overall flow organization; second, distinct phase-dependent behavioral transitions exist between the main tourist corridor and lateral passages, making single origin–destination models insufficient; third, local congestion and global route choice form mutual feedback loops, where control changes in one passage may trigger load redistribution in adjacent areas.

Existing large-scale event security management practices typically integrate system dynamics, system coupling, and system safety theory, combined with high-risk zone delineation and historical crowd behavior patterns, to formulate a series of control measures [17], [18]. These measures include deploying security personnel at critical locations, erecting temporary barriers for isolation and diversion, implementing one-way passage on key channels, and applying batch release in core areas [19], [20]. For instance, during major holidays, the Shanghai Bund area adopts batch release, passage direction regulation, south-entry north-exit routing, and reduction of head-on conflicts [21]; similarly, Nanluoguxiang main street in Beijing implements one-way passage measures (south entry, north exit) during peak periods [22], consistent in control mechanism with the Shanghai Bund approach.

However, current management measures for urban high-pedestrian-flow areas still rely considerably on manual experience and static contingency plans, with deficiencies remaining in scientific evaluation and refined optimization [23]. First, many control schemes are formulated based on historical experience, lacking quantitative characterization of the impact of passage direction settings, entrance release capacities, and phase-dependent behavioral preferences. Second, existing methods tend to focus on single-point statistics or local density monitoring, making it difficult to describe the dynamic coupling among "passage direction rules–entrance flow control–local optimal directions–collective density evolution–exit load distribution" [24]. Third, in open scenic area scenarios with multiple entrances, passages, and stages, neglecting visitor route preferences and phase transition behaviors can easily lead to misjudgment of passage loads and high-risk area locations. Finally, many existing simulation or optimization methods treat the crowd model as a black-box evaluator, failing to exploit the structural information among passage geometry, one-way rules, internal entrance capacities, and multi-route subpopulations [25], resulting in insufficient efficiency in control strategy search [26].

To address the above issues, this paper proposes a macroscopic crowd modeling and control optimization method incorporating passage constraints, fixed geometric guidance, multi-stage route behavior, and internal entrance capacity control, targeting the "waterfront platform–multi-stepped passage" open scenic area scenario. At the modeling level, this paper constructs a Bellman–conservation-law coupled crowd continuum model: different stage–route subpopulations possess independent potential functions with optimal directions computed via the discrete Bellman equation, while sharing the speed–density relationship determined by total density; one-way, bidirectional, and closed passage states are formulated as constraints on the local admissible direction set; and a fixed anisotropic mobility tensor $M(x;\eta_0)$ characterizes geometric channel alignment. We further introduce internal entrance capacity variables $q_c^{\pm}(t)$ to distinguish attempted entrance flow from actually admitted flow, so that excessive demand remains as upstream waiting rather than disappearing from the system. At the optimization level, the controllable variable is $z = (s, q)$, whereas visitor route preferences $\hat{p}$ and the geometric guidance parameter η_0 are fixed model inputs. The scalarized objective combines total travel time, high-density exposure, realized channel-load imbalance, entrance waiting, and capacity-control smoothness. A hierarchical constrained mixed-variable black-box optimization framework (HCMBO) is designed to solve this direction-capacity problem under limited high-fidelity simulation budgets.

The main contributions of this paper are as follows.

1) A "waterfront platform–multi-stepped passage" crowd flow modeling framework for open scenic areas is pro-

posed, representing visitor behavior as a multi-stage, multi-route continuum flow process, where different subpopulations possess independent potential functions while sharing collective congestion feedback.

2) A Bellman–conservation law coupled model integrating passage constraints and geometric guidance is constructed. The admissible direction set first encodes one-way, bidirectional, and closed channel states, while an anisotropic mobility tensor simulates pre-alignment, smooth convergence, and streamline reorganization near passage entrances.
3) An evaluation framework distinguishing mechanism verification from strategy optimization is established, categorizing indicators into behavioral mechanism indicators (direction consistency, approach angle distribution, channel capture domain) and management optimization indicators (total travel time, high-density exposure time, realized channel-flow variance, entrance waiting, and control smoothness).
4) A hierarchical constrained mixed-variable black-box optimization framework is designed for the controllable direction–entrance-rate variable $z = (s, q)$ under fixed historically identified preference parameters and fixed geometric guidance. The final HCMBO mainline concentrates the budget on direction-wise structured entrance-rate optimization and high-fidelity rechecking. The framework is evaluated against random search, pure simulated annealing, enumeration plus differential evolution, and TPE-based mixed-variable Bayesian optimization.

II. Related Work

A. Massive Crowd Management and Control

Crowd management in urban centers has been extensively studied from control-theoretic and optimization perspectives, aiming to enhance safety and efficiency through boundary regulation, facility layout, and intelligent decision-making [27], [28].

In control-theoretic approaches, Zhong et al. [29] designed boundary control strategies for urban traffic flow networks to ensure convergence to uncongested states under disturbances, with stability analysis based on Lyapunov methods [30]. For one-dimensional crowd dynamics, Wadoo and Kachroo [31] employed advection and diffusion control to prevent blocking and shock waves during evacuation. Addressing multi-directional disturbance propagation, Qin [32] proposed a unit sliding mode controller based on integral barrier Lyapunov functions to ensure boundedness of internal state variables, demonstrating strong disturbance rejection capabilities. For heterogeneous corridor regulation, Zhu et al. [33] combined policy learning with neural networks to learn optimal feedback controllers that adjust inflow rates and free-flow speeds, achieving balanced density distribution and reduced congestion.

Physical infrastructure regulation has also proven effective. Studies have shown that strategically placed obstacles can significantly improve outflow, velocity, and local density in merging areas [23], [34]. Yang et al. [35], [36] employed Model Predictive Control (MPC) to optimize barrier lengths at subway bottlenecks, later extending control variables to entrance limiters, gates, and escalator directions for dynamic, real-time crowd regulation. For evacuation guidance, Carmona and Paricio-Garcia [37] proposed dynamic exit-choice recommendations using multinomial Logit models. However, their approach relies on discrete speed instructions (e.g., "stop," "walk," "run") transmitted via electronic wristbands [38], which poses implementation challenges in open urban scenarios.

Recently, intelligent optimization and deep reinforcement learning have emerged as promising alternatives. Liao et al. [39] formulated fence layout as a subset selection problem, while subsequent work [26] proposed a dynamic crowd inflow control system using LSTM networks to capture temporal features and Proximal Policy Optimization (PPO) [40] for real-time entrance flow regulation. Differential Evolution (DE) [41] has also been applied to optimize guardrail layouts and flow guidance in real-world scenarios such as Chengdu East Railway Station, using average travel time and crowd pressure as multi-objective criteria [42].

B. Macroscopic Crowd Modeling

Macroscopic crowd models treat pedestrian flows as continuous media, offering computational efficiency and analytical tractability compared to microscopic approaches. Henderson [43] first applied fluid dynamics to pedestrian traffic in the early 1970s, comparing crowd motion to the behavior of gas or fluid particles.

Hughes [44] pioneered the first-order continuum theory for pedestrian flow, introducing the classical model in which pedestrian walking speed is governed by local density through a fundamental diagram, while movement direction follows the steepest-descent path of a potential function representing the remaining travel time to exits. This framework elegantly captures the interplay between congestion and route choice. Ling et al. [45] subsequently generalized the Hughes formulation and developed an efficient numerical solution combining a weighted essentially non-oscillatory (WENO) scheme for the conservation law with a fast sweeping method for the eikonal equation.

Several extensions have enriched the descriptive capability of macroscopic models. Nonlocal flux models were proposed by Colombo et al. [46], replacing pointwise density with a neighborhood-averaged quantity in the velocity function to account for anticipatory behavior. This direction was further extended to incorporate anisotropic perception fields and boundary effects from walls and obstacles, with high-resolution numerical schemes and well-posedness results suitable for realistic architectural configurations [47]. To enforce the hard incompressibility constraint—that pedestrian density cannot exceed a physically admissible maximum—Maury et al. [48] introduced the projection model, which formulates the actual velocity as a least-squares projection of the desired velocity onto the set of feasible (non-overlapping) configurations. This framework was later recast as a Wasserstein gradient flow with density constraints [49], using the Jordan–Kinderlehrer–Otto (JKO) scheme [50] to discretize the motion while guaranteeing that density respects an upper bound, making it well-suited for high-density congestion scenarios.

To overcome non-physical behaviors such as supersonic information propagation that may arise in first-order models, Aw and Rascle [51] introduced a second-order model based on a generalized momentum conservation law that preserves anisotropy. The Aw–Rascle (AR) framework was subsequently adapted to pedestrian flows [52], where the congestion pressure term successfully reproduces characteristic phenomena including stop-and-go waves and spontaneous lane formation.

Beyond the classical Hughes-type framework, extensions have incorporated control and guidance mechanisms relevant to crowd management. Anisotropic formulations using mobility tensors have been developed to represent channel geometries and directional preferences [53], [54], allowing for continuous geometric guidance without explicit prohibition. Constrained Hamilton-Jacobi-Bellman (HJB) formulations have been employed to model strict directional constraints such as one-way corridors, with the Bellman optimality principle providing a natural framework for unifying anisotropic geometric guidance with discrete directional constraints.

Despite these advances, existing approaches often treat control parameters as fixed or manually tuned. The joint optimization of discrete channel direction configurations and internal entrance capacity profiles remains underexplored, particularly for large-scale urban pedestrian zones with multiple behavioral stages and heterogeneous route choices.

III. Methodology

A. Problem Definition and Overall Framework

This paper addresses a multi-channel crowd management problem in open scenic areas. The proposed method combines a Bellman–conservation-law crowd dynamics model with a joint direction-capacity control formulation. The controllable management variable is defined as

$$z = (s, q) \in \mathcal{Z}, \tag{1}$$

where s denotes the discrete direction states of the channels, and q denotes the time-varying internal entrance capacity profile. The behavioral preference parameter $\hat{p}$ and the geometric guidance parameter η_0 are treated as fixed model inputs. For a given control variable z, the lower-level simulator returns density fields, velocity fields, entrance waiting, and realized channel throughputs. The upper-level optimizer then evaluates efficiency, safety, load balance, waiting, and smoothness metrics, and searches for a better control policy under a limited simulation budget. The overall method framework is shown in Fig. 2.

The key modeling assumption is that direction rules and entrance capacities should be optimized jointly. The direction configuration s changes the local feasible direction set, whereas the entrance capacity q changes the realized admitted flux at channel entrances. These two controls jointly affect channel loading, local high-density exposure, and upstream waiting. Therefore, the paper formulates crowd management as a mixed-variable simulation-based optimization problem rather than as two independent subproblems.

B. Bellman–Conservation-Law Crowd Dynamics

1) Conservation Law and Speed Function: We consider a bounded two-dimensional domain $\Omega \subset \mathbb{R}^2$ representing the pedestrian walkway. Let $\Omega_w \subset \Omega$ denote the walkable region, $\rho(x,t)$ the total crowd density, and $\mathbf{v}(x,t)$ the local mean velocity. Mass conservation is governed by

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \tag{2}$$

The speed magnitude is determined by local congestion. We adopt the Greenshields-type speed function

$$f(\rho) = v_{\max} \left(1 - \frac{\rho}{\rho_{\max}}\right)_+ , \tag{3}$$

where $v_{\max}$ is the free-flow speed, $\rho_{\max}$ is the maximum admissible density, and $(\cdot)_+$ denotes nonnegative truncation. This function captures the feedback that pedestrians move close to free-flow speed in low-density regions and slow down as density approaches congestion.

The classical Hughes model assumes that all movement directions are feasible and isotropic. This assumption is insufficient for crowd management scenarios involving one-way rules, temporary closures, entrance metering, and channel-axis alignment. The following subsections introduce admissible direction sets, fixed anisotropic mobility tensors, and internal entrance capacity control into the Bellman–conservation-law framework.

2) Multi-Stage Multi-Route Density Decomposition: Visitors in open scenic areas usually experience several behavioral stages, such as entry, touring, exit, and return. To represent this behavior chain, the total density is decomposed into stage–route subpopulations:

$$\rho(x,t) = \sum_{\sigma=1}^{S} \sum_{r=1}^{R_\sigma} \rho_{\sigma,r}(x,t), \tag{4}$$

where σ denotes the behavioral stage and r denotes the route type within that stage. All subpopulations share the speed function $f(\rho)$ determined by the total density, but they have different target regions, potential functions, and velocity directions.

For subpopulation (σ, r), the density evolves according to

$$\frac{\partial \rho_{\sigma,r}}{\partial t} + \nabla \cdot (\rho_{\sigma,r} \mathbf{v}_{\sigma,r}) = Q^{\mathrm{in}}_{\sigma,r} - Q^{\mathrm{out}}_{\sigma,r}. \tag{5}$$

If pedestrians in subpopulation (σ, r) complete the current stage in target region $G_{\sigma,r}$ and switch to next-stage route k with probability $\hat{p}_{(\sigma,r)\to k}$, the transition term is

$$Q_{(\sigma,r)\to(\sigma+1,k)}(x,t) = \hat{p}_{(\sigma,r)\to k} \kappa_{\sigma,r} \chi_{G_{\sigma,r}}(x) \rho_{\sigma,r}(x,t), \tag{6}$$

where $\kappa_{\sigma,r}$ is the stage-switching rate and $\chi_{G_{\sigma,r}}$ is the indicator function of the completion region. The parameter $\hat{p}$ represents historically inferred or scenario-given route preferences and is fixed in the optimization problem.

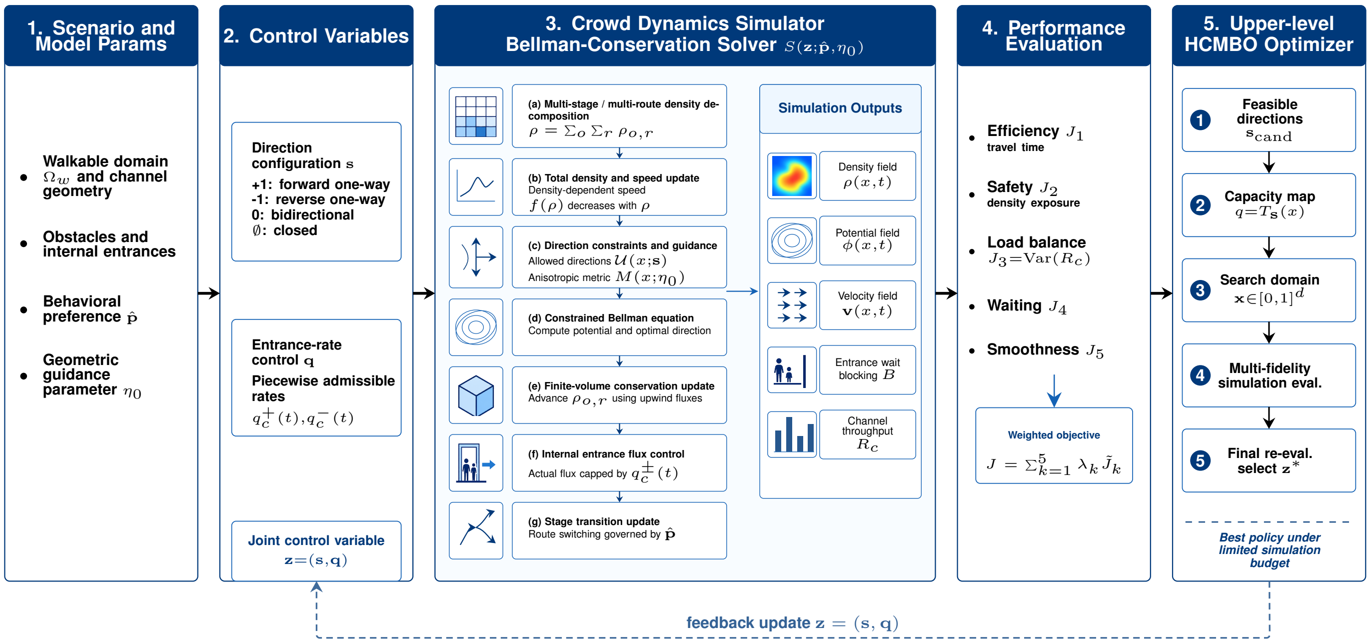

Fig. 2. Method framework for joint direction and entrance-rate optimization. Fixed scenario inputs, behavioral preferences, and geometric guidance parameters define the lower-level Bellman–conservation-law crowd simulator. The controllable direction configuration and entrance-rate profile are jointly evaluated by efficiency, safety, load-balance, waiting, and smoothness objectives, and are updated by the upper-level HCMBO optimizer.

C. Direction Constraints and Geometric Guidance

1) Direction Configuration Variable: Let the direction state of channel c be

$$s_c \in \{+1, -1, 0, \varnothing\}, \tag{7}$$

where $+1$, -1, 0, and $\varnothing$ denote reference forward one-way, reference reverse one-way, bidirectional, and closed states, respectively. If $\tau_c(x)$ is the reference tangent direction of channel c, the local admissible direction set in the channel region is defined as

$$U_c(x; s_c) = \begin{cases} \{\tau_c(x)\}, & s_c = +1, \\ \{-\tau_c(x)\}, & s_c = -1, \\ \{\tau_c(x), -\tau_c(x)\}, & s_c = 0, \\ \varnothing, & s_c = \varnothing, \end{cases} \quad x \in \Omega_c. \tag{8}$$

Thus, s_c changes the local feasible direction set. A closed channel has no admissible direction, a one-way channel retains only one reference tangent, and a bidirectional channel retains both opposite tangential directions.

2) Fixed Anisotropic Mobility Tensor: The admissible set $U(x;s)$ represents hard feasibility, but it does not capture the soft geometric guidance induced by narrow channels. Pedestrians approaching a passage often pre-align with the channel axis before entering it. To represent this effect, a fixed anisotropic mobility tensor $M(x;\eta_0)$ is introduced. Equivalently, it is the inverse of a local travel-cost metric. For channel c, let τ_c and n_c denote its tangent and normal directions. The mobility tensor is defined as

$$M_c(x; \eta_0) = \beta_c \left(\eta_0\, \tau_c \tau_c^\top + n_c n_c^\top\right), \qquad \eta_0 \geq 1, \tag{9}$$

where β_c is a channel weight and η_0 controls the mobility preference strength along the channel axis. Larger η_0 increases tangential mobility, equivalently reducing the effective cost of tangential motion and promoting upstream pre-alignment. In this paper, η_0 is fixed and is not optimized as a control variable.

3) Constrained Bellman Equation: Given $U(x;s)$ and $M(x;\eta_0)$, the potential function $\phi(x,t)$ represents the minimum remaining cost to the corresponding target region. The discrete Bellman update is

$$\phi(x) = \min_{\mathbf{u}\in U(x;s)} \left[\phi(x + \Delta x\, \mathbf{u}) + \frac{\Delta x}{f(\rho(x,t))} \frac{1}{\sqrt{\mathbf{u}^\top M(x;\eta_0)\mathbf{u}}} \right]. \tag{10}$$

The optimal direction is

$$\mathbf{u}^*(x,t) = \arg\min_{\mathbf{u}\in U(x;s)} (\cdots), \tag{11}$$

and the velocity is

$$\mathbf{v}(x,t) = f(\rho(x,t))\, \mathbf{u}^*(x,t). \tag{12}$$

For the multi-stage multi-route model, each subpopulation (σ, r) uses its own target-specific potential $\phi_{\sigma,r}$ while sharing the same total-density congestion feedback. Thus, direction rules, geometric guidance, and density-dependent congestion are coupled in a single Bellman–conservation-law model.

D. Internal Entrance Capacity Control

Direction rules alone cannot represent the management action in which a channel is open but its entrance must be metered. For each channel, we introduce maximum internal entrance rates

$$q = \{q_c^+(t), q_c^-(t)\}_{c=1}^{C}, \tag{13}$$

where $q_c^+(t)$ and $q_c^-(t)$ denote the maximum admitted rates at the reference forward and reverse entrances of channel c. These controls act on internal channel entrance fluxes rather than on external boundary inflows.

The direction state and entrance capacity must satisfy the hard consistency constraints

$$\begin{aligned} s_c = +1 &\Rightarrow q_c^-(t) = 0, \\ s_c = -1 &\Rightarrow q_c^+(t) = 0, \\ s_c = \varnothing &\Rightarrow q_c^+(t) = q_c^-(t) = 0, \\ s_c = 0 &\Rightarrow q_c^+(t) + q_c^-(t) \leq \bar{q}_c. \end{aligned} \tag{14}$$

Let $A_c^\pm(t)$ be the attempted entrance flux induced by free movement. The actually admitted flux is

$$\hat{A}_c^\pm(t) = \min\{A_c^\pm(t), q_c^\pm(t)\}. \tag{15}$$

When $A_c^\pm(t) > 0$, the flux across the internal entrance is scaled by

$$\theta_c^\pm(t) = \frac{\hat{A}_c^\pm(t)}{A_c^\pm(t)}. \tag{16}$$

The excess demand $(A_c^\pm(t) - \hat{A}_c^\pm(t))_+$ remains upstream and is recorded as waiting or queueing mass. This treatment changes channel usage and risk distribution while preserving the mass balance of the conservation-law update.

E. Numerical Solver and Simulation Operator

The walkable region, obstacles, channel areas, and internal entrance interfaces are discretized on a uniform two-dimensional grid. At each time step, the total density is first computed from all stage–route subpopulations and used to update $f(\rho)$. For each subpopulation, the admissible direction set and fixed mobility tensor are then constructed, and the corresponding discrete Bellman equation is solved. After the velocity direction is recovered, an upwind finite-volume update advances the conservation law, while internal entrance fluxes are modified by $\theta_c^\pm(t)$. Finally, stage transition terms are applied to update the subpopulation densities.

For any feasible control variable $z = (s, q)$, the complete numerical workflow defines a lower-level simulation operator

$$(\rho, \phi, \mathbf{v}, B, R) = \mathcal{S}(z; \hat{p}, \eta_0), \tag{17}$$

where B records entrance waiting or blocking and $R = (R_1, \ldots, R_C)$ records realized cumulative throughputs of the channels. The channel balance metric below is computed from the realized throughputs R_c, not from the prescribed capacity limits $q_c^\pm(t)$.

F. Evaluation Metrics and Optimization Objective

1) Efficiency, Safety, and Load Balance: The total travel time metric is

$$J_1(z) = \int_0^T \int_{\Omega_w} \rho(x, t)\, dx\, dt. \tag{18}$$

The high-density exposure metric is

$$J_2(z) = \int_0^T \int_{\Omega_w} \mathbf{1}_{\rho(x,t) > \rho_{\text{safe}}}\, dx\, dt. \tag{19}$$

Let

$$R_c(T) = \int_0^T (\hat{A}_c^+(t) + \hat{A}_c^-(t))\, dt \tag{20}$$

be the realized cumulative throughput of channel c. The channel load imbalance is

$$J_3(z) = \mathrm{Var}(R_1(T), \ldots, R_C(T)). \tag{21}$$

2) Entrance Waiting and Control Smoothness: Entrance waiting is penalized as

$$J_4(z) = \int_0^T \sum_{c=1}^{C} \left(B_c^+(t) + B_c^-(t)\right)\, dt. \tag{22}$$

For piecewise-constant capacity profiles, smoothness is measured by

$$J_5(z) = \sum_{c,k} \left[(q_{c,k+1}^+ - q_{c,k}^+)^2 + (q_{c,k+1}^- - q_{c,k}^-)^2\right]. \tag{23}$$

Here k indexes the capacity-control time segment. J_4 discourages excessive entrance metering that creates upstream queues, while J_5 discourages abrupt high-frequency changes in capacity profiles.

3) Standardized Scalar Objective: Because the evaluation terms have different units and magnitudes, they are standardized before weighting:

$$\tilde{J}_1, \quad \tilde{J}_2, \quad \tilde{J}_3, \quad \tilde{J}_4, \quad \tilde{J}_5.$$

Given weights $\lambda = (\lambda_1, \ldots, \lambda_5)$, the scalarized objective is

$$J(z) = \sum_{k=1}^{5} \lambda_k \tilde{J}_k(z), \tag{24}$$

and the optimization problem is

$$\min_{z \in \mathcal{Z}} J(z), \qquad z = (s, q). \tag{25}$$

The paper solves this fixed-weight scalarized management problem. Different weights can represent different management preferences, but the algorithm itself does not depend on one specific objective term being independently optimal.

G. Hierarchical Constrained Mixed-Variable Black-Box Optimization

Since s is discrete, q is continuous or piecewise continuous, and each objective evaluation requires the expensive simulator $\mathcal{S}$, the resulting problem is an expensive mixed-variable black-box optimization problem. This paper proposes HCMBO, a Hierarchical Constrained Mixed-variable Black-box Optimization method, to exploit the structural relation between direction variables and capacity variables.

1) Direction Candidates and Capacity Parameterization: HCMBO first generates a direction candidate set $\mathcal{S}_{\text{cand}}$ satisfying operational and connectivity constraints. For a fixed direction configuration s, the entrance capacity profile is represented by a low-dimensional continuous vector $x \in [0, 1]^d$ and mapped to the actual capacity control by

$$q = T_s(x), \qquad x \in [0, 1]^d. \tag{26}$$

TABLE I
MAIN WORKFLOW OF HCMBO

Step	Operation
1	Generate the direction candidate set $\mathcal{S}_{\mathrm{cand}}$ and remove configurations violating operational or connectivity constraints.
2	For each direction s, construct the direction-aware capacity map $q = T_s(x)$.
3	Perform structured black-box search in the capacity parameter space $x \in [0,1]^d$.
4	Evaluate candidates with $\mathcal{S}(z; \hat{p}, \eta_0)$ and update surrogate information.
5	Select near-optimal candidates for unified high-fidelity rechecking.
6	Output the final policy z^* according to the high-fidelity objective value.

The mapping T_s automatically applies direction masks, capacity upper bounds, closure constraints, and smoothness restrictions. For example, if a channel is set to reference forward one-way, the reverse entrance capacity is set to zero; if a channel is closed, both directional capacities are set to zero. Thus, the optimizer searches in the continuous x-space while all candidates are mapped to capacity profiles consistent with the discrete direction configuration.

2) Structured Capacity Search: For each direction candidate s, HCMBO performs structured search in the corresponding capacity parameter space. The search uses previously evaluated samples to construct surrogate information for objective values and feasibility, and preferentially selects capacity candidates that satisfy constraints and may improve the incumbent objective. This hierarchical design avoids blind sampling in the full mixed-variable space and separates discrete direction structure from continuous capacity structure.

Multi-fidelity evaluation can be used during search. Low- or medium-fidelity simulations help identify clearly poor capacity profiles, whereas high-fidelity simulations are reserved for final candidate rechecking and ranking. To avoid discarding good long-horizon strategies due to short-horizon bias, low-fidelity results are used only as auxiliary evaluation signals and do not directly determine the final optimum.

3) High-Fidelity Rechecking and Final Selection: All finalist candidates are re-evaluated under a unified high-fidelity setting. Let $\mathcal{C}_{\mathrm{HF}}$ be the candidate set selected for high-fidelity rechecking. The final output is

$$z^* = \arg\min_{z \in \mathcal{C}_{\mathrm{HF}}} J_{\mathrm{HF}}(z). \tag{27}$$

This mechanism prevents surrogate error, low-fidelity bias, or intermediate search ranking from directly becoming the final performance claim.

The main workflow of HCMBO is summarized in Table I.

Compared with generic mixed-variable optimization, HCMBO explicitly exploits direction-capacity consistency. The direction variable determines the effective feasible structure of the capacity variable, and the capacity variable determines the continuous response of entrance release, waiting, and channel load. This structured treatment reduces invalid candidates and concentrates the finite simulation budget on feasible policies with clear management meaning.

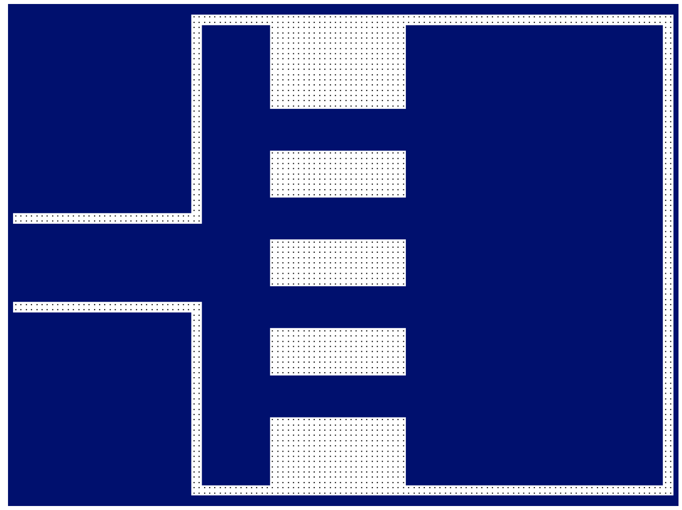

Fig. 3. Four-channel scenic-platform simulation scenario used in the numerical experiments.

IV. Numerical Experiments

This section validates the proposed model and optimizer on a four-channel scenic-platform scenario abstracted from a waterfront viewing area. The scenario consists of an inflow region, a viewing and exit region, a central wall, and four connecting passages: upper, middle, lower-middle, and bottom passages. The experiments follow the same logic as the Chinese experimental outline: mechanism validation, directional-response analysis, entrance-rate response, horizontal optimizer comparison, and HCMBO ablation.

A. Experimental Setup

The domain is discretized on a 128×96 grid with spacing $\Delta x = 0.5$. The passage width is 2, the controlled internal entrance interface width is 8, the free-flow speed is $v_{\max} = 1.5$, the maximum density is $\rho_{\max} = 5.0$, the initial density is $\rho_{\mathrm{init}} = 2.2$, and the CFL factor is 0.9. Mechanism and response experiments use the corresponding validation horizons, whereas the optimizer comparison uses high-fidelity rechecks with 1600 simulation steps, horizon 160, and Bellman updates every five steps. Final optimizer rankings are based on the high-fidelity objective in Section III-F. A high-fidelity best candidate is called feasible if the density-cap removal mass is no more than 2% of the reference mass.

B. Mechanism Validation

The first experiment verifies whether the two local modeling components have the intended behavioral meanings. The fixed anisotropic mobility tensor $M(x; \eta_0)$ should reshape the channel attraction domain and induce entrance pre-alignment, whereas the admissible direction set $U(x; s)$ should remove prohibited directions and reroute flow through feasible passages. When M is applied to the middle channel, its flow share

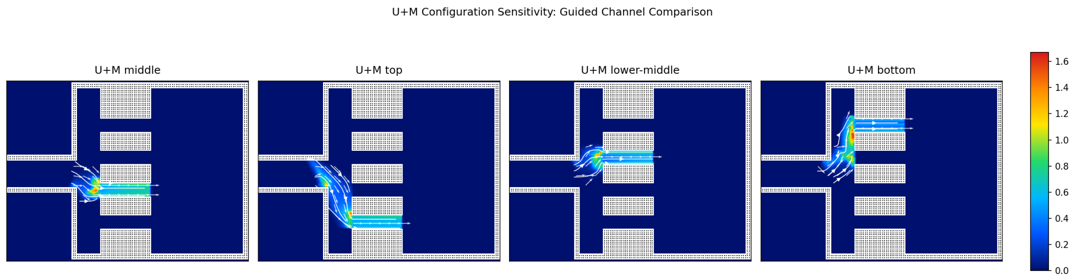

Fig. 4. Spatial transferability of the combined $U+M$ mechanism. Moving the controlled channel induces corresponding migration of dominant flow channels, streamlines, and density hotspots.

TABLE II
NON-DOMINATED DIRECTIONAL-CONTROL CASES

Case	Policy	Code	$\tilde{J}_1$	$\tilde{J}_2$	$\tilde{J}_3$	$\tilde{J}$	Main characteristic
C1	BSL	FFFF	0.3535	**0.00785**	0.1531	0.5145	Lowest exposure
C3	SE-M	WEWW	**0.3141**	0.01680	0.2741	0.6049	Best efficiency but risk-concentrated
C4	SE-LM	WWEW	0.3335	0.01662	0.3131	0.6632	Efficient but load-imbalanced
C6	SR-T	WEEE	0.4029	0.01174	0.1335	0.5482	High return throughput
C8	SR-LM	EEWE	0.3395	0.01311	0.1576	**0.5102**	Lowest aggregate objective
C9	SR-B	EEEW	0.4377	0.01344	**0.1138**	0.5649	Most balanced channel loads

increases from 48.78% in the baseline to 100.00%. When a one-way rule excludes the originally dominant middle-channel direction, the middle-channel share becomes zero and the flow is redistributed to the upper, lower-middle, and bottom channels with shares 19.25%, 68.76%, and 12.00%, respectively. In a bidirectional inflow test, imposing an eastbound one-way rule reduces the westbound middle-channel share, reverse-direction mass time, and counterflow cell time from 73.18%, 415.75, and 1213.76 to zero.

Fig. 4 further shows that the combined $U+M$ mechanism is spatially transferable. Changing the guided channel from the upper to the lower passage moves the dominant flow channel and local density hotspot accordingly. These results support using the Bellman–conservation-law simulator as the lower-level evaluator for subsequent strategy optimization.

C. Directional-Control Response

We next examine whether the discrete direction configuration s creates nontrivial system-level trade-offs. The response experiment uses thirteen interpretable direction settings: an all-free baseline, four single-entry cases, four single-return cases, and four one-channel-closure cases. The direction code is ordered as $[T, M, LM, B]$, where E, W, F, and X denote eastbound entry, westbound return, bidirectional free passage, and closure.

Fig. 5 and Table II show that no single direction configuration dominates all objectives. C8 (SR-LM, EEWE) obtains the lowest aggregate value $\tilde{J} = 0.5102$, C1 gives the lowest safety exposure $\tilde{J}_2 = 0.00785$, C3 gives the best efficiency term $\tilde{J}_1 = 0.3141$, and C9 gives the best load-balance term $\tilde{J}_3 = 0.1138$. This non-monotonic response motivates automatic mixed-variable optimization rather than manual selection from a small set of rules.

D. Entrance-Rate Response

Direction rules alone cannot represent the case in which a channel remains open but its internal entrance must be metered. We therefore fix $[T, M, LM, B] = [F, E, W, F]$ and scan four internal entrance-rate levels: no limit, high, medium, and low. The rate upper bound acts on internal entrance crossing flow. When attempted flow exceeds the upper bound, only the admitted flow crosses the interface, and the unreleased attempted flow remains upstream as waiting mass. The entrance binding-time ratio is defined as the fraction of time steps in which attempted entrance flow exceeds the rate upper bound and creates unreleased attempted flow.

Table III and Fig. 6 show that tightening the rate upper bound does not simply improve safety. From the no-limit case to the low level, unreleased attempted flow increases from 0 to 427.89, the binding ratio increases from 0 to 0.721, and $\tilde{J}_2^{\mathrm{eval}}$ increases from 0.174 to 0.466. Thus, q is a genuine continuous control variable that changes system response and must be optimized jointly with s.

E. Horizontal Optimization Comparison

The horizontal comparison evaluates Prior Baseline, Random Search, Pure Simulated Annealing, Enumeration plus Differential Evolution, TPE-Mixed BO, and HCMBO. All optimizers use $B = 400$ candidate evaluations and top-10 high-fidelity rechecks over seeds 11, 23, 37, 51, and 73. The final HCMBO data are taken from the G7-D mainline, whereas the other baselines are taken from the G6 horizontal comparison under the same high-fidelity protocol.

Table IV shows that HCMBO achieves the lowest mean objective, lowest median objective, lowest best objective, and highest feasible rate. Relative to TPE-Mixed BO, HCMBO is better in $4/5$ paired seeds, with an average objective difference of -0.1066. It also outperforms Random Search in all five

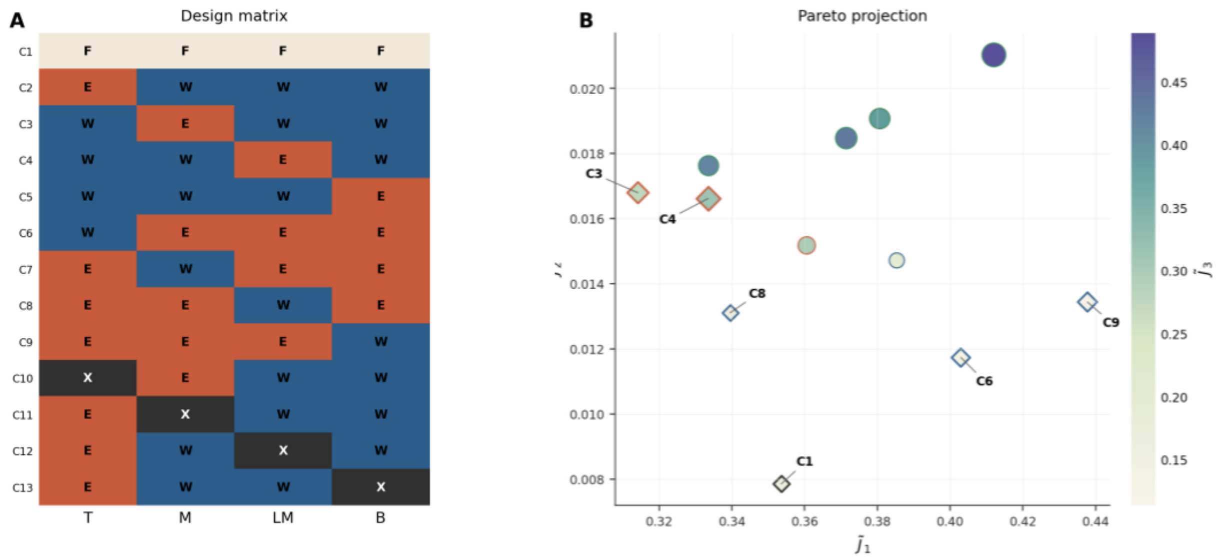

Fig. 5. Directional-control design matrix and objective-space trade-off. Color encodes $\tilde{J}_3$, marker size encodes $\tilde{J} = \tilde{J}_1 + \tilde{J}_2 + \tilde{J}_3$, and diamonds indicate non-dominated cases.

TABLE III
RESPONSE UNDER INTERNAL ENTRANCE-RATE UPPER-BOUND LEVELS

Level	$\tilde{J}_1$	$\tilde{J}_2^{\mathrm{eval}}$	$\tilde{J}_3$	Objective	Unreleased attempted flow	Binding ratio
No limit	0.6274	0.174	0.8838	1.6851	0.00	0.000
High	0.6262	0.358	0.8814	1.8652	241.40	0.605
Medium	0.6228	0.443	0.8620	1.9276	329.26	0.666
Low	0.6181	0.466	0.8493	1.9337	427.89	0.721

TABLE IV
HIGH-FIDELITY PERFORMANCE IN THE DIRECTION–ENTRANCE-RATE OPTIMIZATION COMPARISON

Method	Mean objective	Std.	Median	Best	Worst	Feasible rate
HCMBO	**2.6338**	0.3106	**2.5065**	**2.4495**	3.1851	**100%**
TPE-Mixed BO	2.7403	0.2475	2.6288	2.5180	3.1345	80%
Pure SA	2.9248	0.1386	2.9203	2.7468	3.1049	60%
Random Search	3.0112	0.1319	2.9797	2.8821	3.2287	40%
Enum-DE	3.0597	0.1932	3.1004	2.7329	3.2500	40%
Prior Baseline	5.4774	0.0000	5.4774	5.4774	5.4774	0%

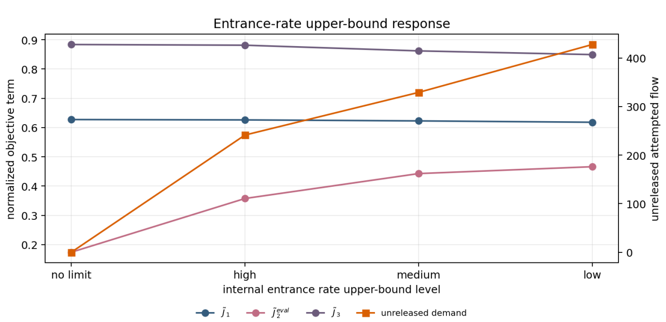

Fig. 6. Objective-term response under internal entrance-rate upper-bound levels. Tightening the upper bound changes safety exposure, load balance, and unreleased attempted flow.

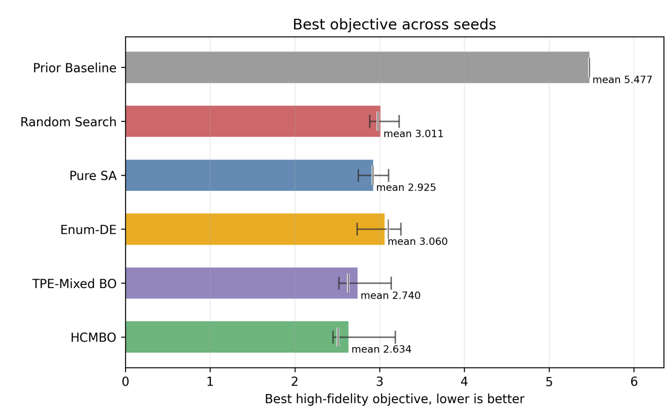

Fig. 7. Best high-fidelity objective across seeds. Range bars show the seed-wise spread, and the short vertical mark indicates the median.

seeds and outperforms Pure SA and Enum-DE in four of five seeds. The remaining high objective in one HCMBO seed is mainly driven by a larger safety exposure term, indicating a residual need for stronger safety-constraint awareness.

Figs. 7–9 provide complementary visual evidence. TPE-Mixed BO descends quickly in the early evaluations, but

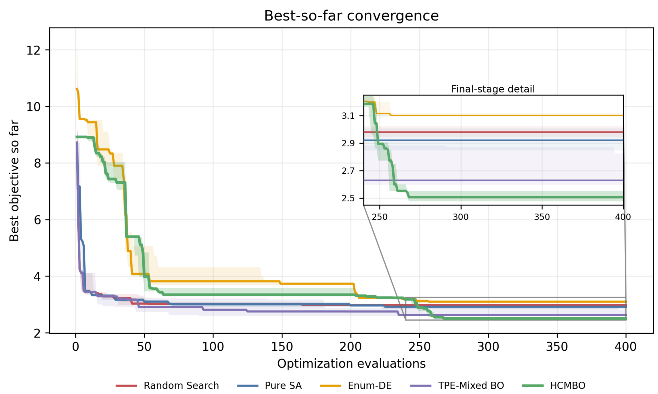

Fig. 8. Best-so-far convergence curves under the same 400-evaluation budget. The inset highlights the final-stage separation among methods.

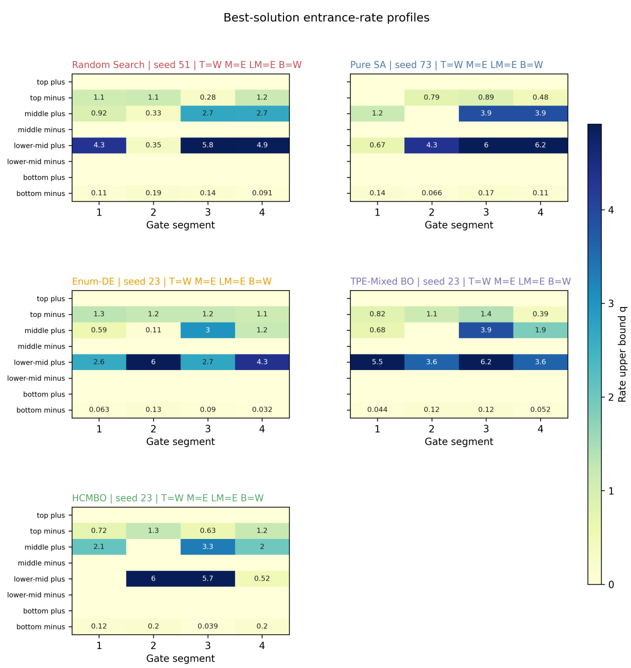

Fig. 9. Best-solution entrance-rate profiles for Random Search, Pure SA, Enum-DE, TPE-Mixed BO, and HCMBO. The common direction structure is $[T, M, LM, B] = [W, E, E, W]$ for several methods, but the segment-wise rate allocations differ substantially.

HCMBO continues to improve in the middle and late stages and reaches the lowest final region. The entrance-rate profiles show that performance differences are not only caused by choosing a discrete direction code; they also arise from detailed segment-wise rate allocation within the same direction structure.

F. HCMBO Ablation

The ablation experiments examine whether the control variables and search mechanisms used by HCMBO are necessary. First, direction and entrance-rate variables must be optimized jointly. Optimizing only direction with a fixed high rate gives best high-fidelity objective 3.3751, while optimizing only rates under prior directions gives 3.3140; both are clearly worse than joint optimization. Second, short-horizon low-fidelity hard screening increases the best objective to 2.9441 and yields an infeasible best candidate, because early simulations do not reliably reflect long-horizon entrance waiting and load redistribution. Third, removing the entrance-waiting penalty slightly reduces the scalar objective to 2.6015, but increases unreleased attempted flow to 3239.295, showing that waiting should remain in the management objective.

Table V shows that the final HCMBO mainline obtains the lowest scalar objective among the tested variants. Queue-aware LCB and RF-style constrained BO reduce unreleased attempted flow to 2412.404, but their scalar objective increases to 2.6162. Thus, queue information is useful for executable and safety-aware extensions, whereas the final mainline should retain structured direction-wise rate search as the primary budget allocation mechanism.

G. Transfer Validation on a Bund-Inspired Scenario

The preceding experiments use a standardized four-channel platform scenario to verify modeling mechanisms and optimization behavior under controlled conditions. As a supplementary validation, we further construct a simplified abstract scenario inspired by the spatial form of the Shanghai Bund waterfront. This experiment does not claim to reproduce all architectural details of the real site. Instead, it preserves the main waterfront platform, the lateral channel connections, the major inflow location, and the dominant movement direction, so as to examine whether an HCMBO-derived policy can still change the load distribution in a more realistic spatial outline.

The computational grid has resolution 512×224 with $\Delta x = 0.125$. Pedestrians enter from the outer-side entrance, pass through the middle connecting channels, and move toward the waterfront platform and the central platform target region. The inflow-rate multiplier is 3.0, corresponding to an actual inflow rate of 0.24; the maximum density is $\rho_{\max} = 4.0$, and the transition smoothing parameter is $\kappa = 32$. The experimental channels `channel_1`, `channel_2`, `channel_3`, and `channel_4` correspond to the locally named Bund Channels 3, 4, 5, and 6, respectively. Both the controlled and uncontrolled groups use 1600 steps and a time horizon of 50. For the controlled group, the transferred entrance-rate bounds are multiplied by a capacity-scale factor of 3, providing a less restrictive entrance-capacity setting.

The controlled group uses the best HCMBO policy obtained from the multi-direction high-fidelity recheck. Its high-fidelity objective in the optimization stage is 0.7350, and its direction structure leaves the upper and lower channels bidirectional while assigning opposite dominant directions to the two middle channels. Table VI reports the effective two-segment entrance-rate upper bounds after applying the capacity-scale factor of 3.

Table VII compares the controlled and uncontrolled groups under the same high-fidelity protocol. The scalar objective decreases from 1.0140 to 0.7410, corresponding to an absolute reduction of approximately 0.2730 and a relative reduction of 26.9%. The main improvement comes from the load-balance term J_3, which decreases from 0.6043 to 0.2908. This

TABLE V
SEARCH-MECHANISM ABLATION UNDER SEED 23

Variant	Best objective	Feasible rate	J_2	Unreleased flow	Main meaning
HCMBO	**2.4495**	100%	**1.6759**	2545.508	Structured direction-wise rate search
Queue-aware LCB	2.6162	100%	1.8521	**2412.404**	Queue-aware acquisition
RF-style constrained BO	2.6162	100%	1.8521	**2412.404**	Constraint surrogate variant
Adaptive racing	2.6561	100%	1.9474	3204.194	Stage-wise budget concentration
Adaptive racing + queue-aware	2.6683	100%	1.7862	2664.373	Combined racing and queue signal
Trust-region search	3.0317	100%	1.9224	3967.226	Local trust-region search

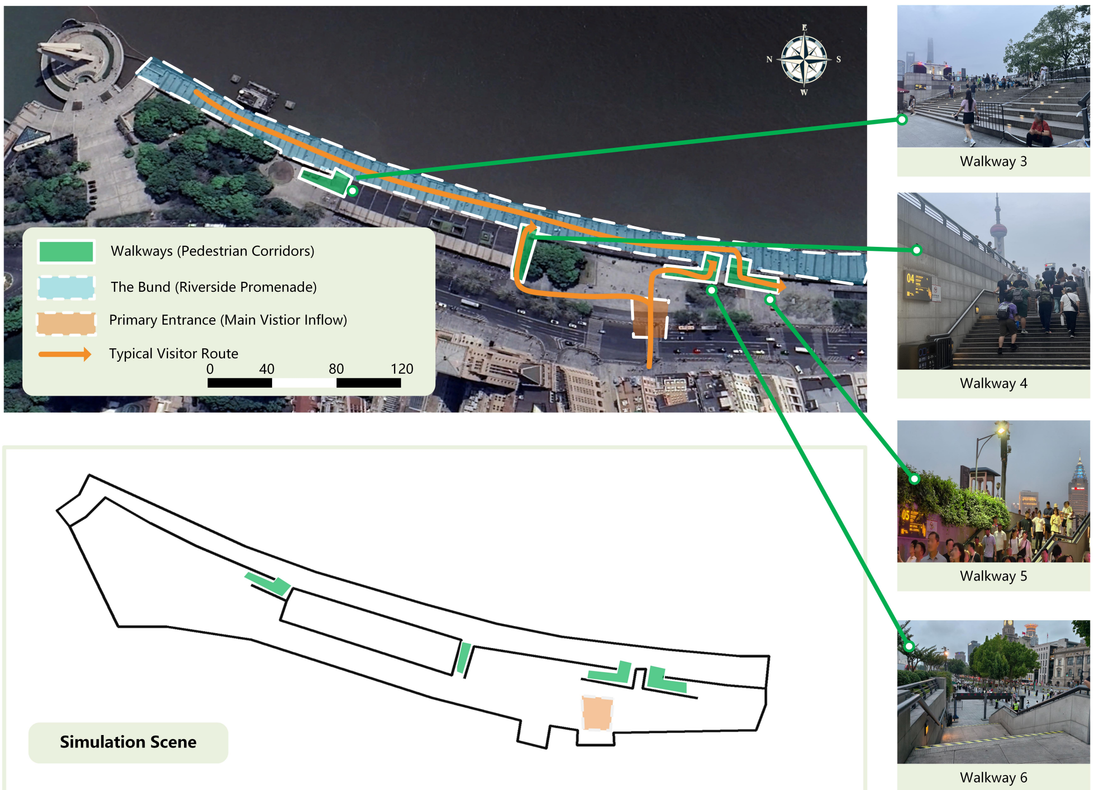

Fig. 10. Real Bund scene and its simplified simulation abstraction. The abstraction retains the waterfront platform boundary, the main entrance, and the channel connections used for control validation.

TABLE VI
EFFECTIVE ENTRANCE-RATE UPPER BOUNDS IN THE BUND TRANSFER EXPERIMENT

Experimental channel	Local channel name	Entrance side	Segment 1	Segment 2
`channel_1`	Bund Channel 3	Forward entrance	0.2474	0.6000
`channel_1`	Bund Channel 3	Reverse entrance	0.2269	0.0000
`channel_2`	Bund Channel 4	Forward entrance	0.6000	0.1688
`channel_2`	Bund Channel 4	Reverse entrance	0.0000	0.0000
`channel_3`	Bund Channel 5	Forward entrance	0.0000	0.0000
`channel_3`	Bund Channel 5	Reverse entrance	0.2560	0.4975
`channel_4`	Bund Channel 6	Forward entrance	0.3785	0.2169
`channel_4`	Bund Channel 6	Reverse entrance	0.2215	0.2122

indicates that the transferred policy changes how pedestrians use the channels, rather than merely reducing all density values uniformly.

The channel-flow shares in Table VIII further explain this reduction. In the uncontrolled group, 82.30% of the channel flow concentrates on `channel_4`, i.e., Bund Channel 6. After applying the HCMBO policy, the flow is redistributed mainly to `channel_2` and `channel_3`, whose combined share reaches approximately 91.34%.

Fig. 11 visualizes the density evolution at steps 50, 350, 900, and 1500. Both groups initially form a local density concentration near the right-side platform target region, reflecting the same inflow and target-attraction structure. At later stages, the uncontrolled case remains more concentrated around the terminal channel and the adjacent platform area, whereas the controlled case develops a more stable flow band through the middle-channel group. This visual evidence is consistent with the channel-share statistics and confirms that the main effect of the HCMBO policy is load redistribution across channels.

The supplementary transfer experiment also reveals an

TABLE VII
CONTROLLED AND UNCONTROLLED RESULTS IN THE BUND TRANSFER EXPERIMENT

Metric	Controlled	Uncontrolled
Objective J	0.7410	1.0140
J_1	0.4145	0.3813
J_2^{eval}	0.0357	0.0284
J_3	0.2908	0.6043
Final cumulative sink flow	6.8237	8.5673
Final cumulative inflow	11.9193	11.9995
Final system mass	7.8455	6.1822
Peak density maximum	≈ 4.0	≈ 4.0
Average mean density	0.003470	0.003210
Attempted entrance flow	7.6454	0
Actual entrance flow	5.7793	0
Unreleased attempted flow	1.8660	0

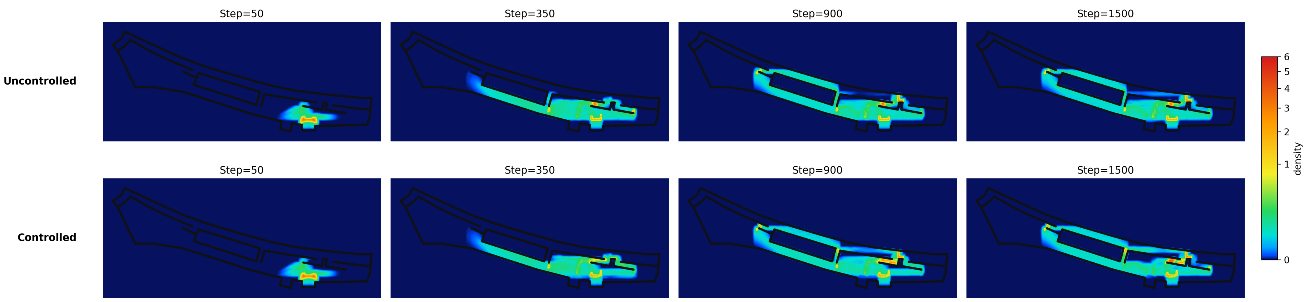

Fig. 11. Density evolution comparison between uncontrolled and controlled cases at steps 50, 350, 900, and 1500. The controlled policy redistributes the later-stage flow from the terminal channel toward the middle-channel group.

TABLE VIII
CHANNEL-FLOW SHARES IN THE BUND TRANSFER EXPERIMENT

Channel	Local name	Controlled	Uncontrolled
`channel_1`	Bund Ch. 3	2.83%	0.32%
`channel_2`	Bund Ch. 4	30.38%	1.35%
`channel_3`	Bund Ch. 5	60.96%	16.03%
`channel_4`	Bund Ch. 6	5.83%	82.30%

important trade-off. The controlled group has a higher efficiency term (J_1 increases from 0.3813 to 0.4145), a slightly higher safety-exposure evaluation term (J_2^{eval} increases from 0.0284 to 0.0357), a smaller cumulative sink flow (6.8237 versus 8.5673), and a larger final system mass (7.8455 versus 6.1822). Therefore, the transferred HCMBO policy should not be interpreted as uniformly improving safety and efficiency. Rather, under the present weights, it reduces the scalar objective mainly by improving channel-load balance, while sacrificing part of the exit throughput and travel efficiency. The capacity-scale factor of 3 reduces unreleased attempted flow to 1.8660, but it does not remove the efficiency cost of the control policy. Overall, the supplementary Bund-inspired experiment shows that the optimized HCMBO policy can still reduce the scalar objective and redistribute flow away from the dominant terminal channel in a simplified real-world spatial abstraction. However, the improvement is accompanied by unreleased attempted entrance flow, lower exit throughput, and higher residual mass in the system. Future deployment-oriented extensions should therefore include explicit throughput constraints, waiting upper bounds, or multi-weight scenario analysis to avoid obtaining policies that improve load balance at the cost of overly strong metering.

V. DISCUSSION

A. Role of the Real-World Scenario Prototype

This study uses the Shanghai Nanjing East Road–Bund area as a real-world prototype for a class of open pedestrian-zone management problems involving multi-channel connections, one-way passage rules, diversion organization, and entrance-rate control. The proposed method does not depend on a complete geometric reconstruction of one specific site. Instead, it abstracts the essential structure of a "viewing platform–multi-channel connection" scenario and builds a macroscopic crowd dynamics model and control optimization framework for this scenario class. Accordingly, the Nanjing East Road–Bund case mainly provides the practical background and modeling motivation, while the simplified numerical scenario is used to verify model mechanisms and optimization behavior under controlled conditions.

B. Interpretation of the Simplified Experiments

The simplified simulation scenario is not intended to reproduce the exact pedestrian flow of a particular time period at the Nanjing East Road–Bund area. Its purpose is to preserve the key structural and operational elements observed in such real settings, including the platform region, connecting passages, channel direction constraints, geometric guidance, entrance release control, and channel-load redistribution. This abstraction makes it possible to examine, in a transparent way, whether the model can represent one-way passage, channel closure, geometric pre-alignment, and multi-stage route transfer. It also enables a fair comparison of different optimization algorithms under the same simulation evaluator. Therefore, the experimental results should be interpreted as evidence of methodological effectiveness and mechanistic interpretability, rather than as a direct operational control plan for the Nanjing East Road–Bund site.

C. Requirements for Field Validation

Real-scene experiments are not conducted in the present study because they require additional data acquisition, model calibration, and field implementation conditions. Practical deployment would require observations of time-varying inflow, local density, velocity fields, channel flow, queue length, and pedestrian route choices. It would also require calibration of the speed–density relationship, route preferences, stage-transition parameters, channel capacities, and entrance-rate upper bounds. Moreover, one-way channel settings, batch release, and on-site guidance measures are tied to staff deployment, temporary facilities, emergency response plans, and safety approval procedures; simulation-based optimization cannot directly replace these operational processes. Future work will combine real geometric data with multi-source pedestrian observations, calibrate the proposed model against field measurements, and incorporate executable management rules into the optimization constraints so that the framework can be extended toward scenario-specific validation and decision support for the Nanjing East Road–Bund area and other open public spaces.

VI. Conclusion

This paper proposed a Bellman–conservation-law crowd management framework for open pedestrian zones with multi-channel direction rules and internal entrance-rate control. The admissible direction set $U(x;s)$ represents hard operational rules such as one-way, bidirectional, and closed passages; the fixed mobility tensor $M(x;\eta_0)$ represents geometric guidance; and the multi-stage route structure describes entry, viewing, exit, and return movements. The controllable variable is $z=(s,q)$, where s is the channel direction configuration and q is the internal entrance-rate upper-bound profile; behavioral preference $\hat{p}$ and geometric guidance strength η_0 are fixed model inputs.

The numerical experiments support three conclusions. First, the lower-level simulator produces interpretable mechanisms: $U(x;s)$ removes prohibited local directions, $M(x;\eta_0)$ changes channel attraction domains and entrance pre-alignment, and the combined $U+M$ mechanism transfers across controlled-channel locations. Second, both control variables are necessary. Directional settings produce non-monotonic efficiency–safety–load-balance trade-offs, and entrance-rate upper bounds significantly change unreleased attempted flow, binding-time ratios, and safety exposure. Third, the final HCMBO optimizer effectively exploits the direction–rate hierarchy. Under five random seeds, $B=400$ candidate evaluations, and unified high-fidelity rechecks, HCMBO obtains the lowest mean objective of 2.6338 and the highest feasible rate of 100%, outperforming TPE-Mixed BO, Random Search, Pure SA, Enum-DE, and the prior baseline. Compared with TPE-Mixed BO, HCMBO is better in $4/5$ paired seeds, with an average objective difference of -0.1066. Ablation results further show that joint direction–rate optimization, removal of low-fidelity hard screening, and structured direction-wise rate search are key contributors to this performance.

The current study still has limitations. The experiments are based on a four-channel abstract scenario, objective weights and finite random seeds affect optimizer ranking, one HCMBO seed still exhibits relatively high safety exposure, and real crowd calibration and online demand updates are not yet included. Future work will extend the framework to more complex real geometries, broader demand levels and random-seed sets, stronger constraint-aware Bayesian optimization or SMAC-style baselines, multi-weight Pareto analysis, and calibration against field observations.

References

[1] P. Charalambous, J. Pettre, V. Vassiliades, Y. Chrysanthou, and N. Pelechano, "Greil-crowds: Crowd simulation with deep reinforcement learning and examples," *ACM Trans. Graph.*, vol. 42, no. 4, p. Article 137, 2023.

[2] H. Jiang, X. Zhang, Y. Dong, and J. Wang, "Data-driven predictive modeling of citywide crowd flow for urban safety management: A case study of beijing, china," *Journal of Forecasting*, vol. 44, no. 2, pp. 730–752, 2025.

[3] D. Shi, J. Li, Q. Wang, J. Chen, R. Lovreglio, J. T. Y. Lo, and J. Ma, "Modeling and simulation of crowd dynamics at evacuation bottlenecks during flood disasters," *Transportation Research Part E: Logistics and Transportation Review*, vol. 198, p. 104064, 2025.

[4] R. Zhao, D. Wang, Y. Wang, C. Han, P. Jia, C. Li, and Y. Ma, "Macroscopic view: Crowd evacuation dynamics at t-shaped street junctions using a modified aw-rascle traffic flow model," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 10, pp. 6612–6621, 2021.

[5] B. L. Zhou, X. O. Tang, and X. G. Wang, "Learning collective crowd behaviors with dynamic pedestrian-agents," *International Journal of Computer Vision*, vol. 111, no. 1, pp. 50–68, 2015.

[6] T. Bosse, M. Hoogendoorn, M. C. A. Klein, J. Treur, C. N. van der Wal, and A. van Wissen, "Modelling collective decision making in groups and crowds: Integrating social contagion and interacting emotions, beliefs and intentions," *Autonomous Agents and Multi-Agent Systems*, vol. 27, no. 1, pp. 52–84, 2013.

[7] H. Liang, L. Yang, J. Du, C.-W. Shu, and S. C. Wong, "Modelling crowd pressure and turbulence through a mixed-type continuum approach," *Transportmetrica B: Transport Dynamics*, vol. 12, no. 1, p. 2328774, 2024.

[8] Z. Shahhoseini and M. Sarvi, "Pedestrian crowd flows in shared spaces: Investigating the impact of geometry based on micro and macro scale measures," *Transportation Research Part B: Methodological*, vol. 122, pp. 57–87, 2019.

[9] N. W. F. Bode, M. Chraibi, and S. Holl, "The emergence of macroscopic interactions between intersecting pedestrian streams," *Transportation Research Part B: Methodological*, vol. 119, pp. 197–210, 2019.

[10] F. Bourdin, "Splitting scheme for a macroscopic crowd motion model with congestion for a two-typed population," *Networks and Heterogeneous Media*, vol. 17, no. 5, pp. 783–801, 2022.

[11] C.-Z. T. Xie, J. Xu, B. Zhu, T.-Q. Tang, S. Lo, B. Zhang, and Y. Tian, "Advancing crowd forecasting with graphs across microscopic trajectory to macroscopic dynamics," *Information Fusion*, vol. 106, p. 102275, 2024.

[12] L. Y. Wang and W. X. Huang, "Visitors' consistent stay behavior patterns within free-roaming scenic architectural complexes: Considering impacts of temporal, spatial, and environmental factors," *Frontiers of Architectural Research*, vol. 13, no. 5, pp. 990–1008, 2024.

[13] X. Yang, R. Zhang, F. Pan, Y. Yang, Y. Li, and X. Yang, "Stochastic user equilibrium path planning for crowd evacuation at subway station based on social force model," *Physica A: Statistical Mechanics and its Applications*, vol. 594, p. 127033, 2022.

[14] J. Zhong, T. Cheng, W. L. Liu, P. Yang, Y. Lin, and J. Zhang, "An evolutionary guardrail layout design framework for crowd control in subway stations," *IEEE Transactions on Computational Social Systems*, vol. 10, no. 1, pp. 297–310, 2023.

[15] P. X. Liu, X. Xiao, J. Zhang, R. H. Wu, and H. L. Zhang, "Spatial configuration and online attention: A space syntax perspective," *Sustainability*, vol. 10, no. 1, 2018.

[16] A. M. Caldeira and E. Kastenholz, “Spatiotemporal tourist behaviour in urban destinations: a framework of analysis,” *Tourism Geographies*, vol. 22, no. 1, pp. 22–50, 2020.

[17] M. Haghani, M. Coughlan, B. Crabb, A. Dierickx, C. Feliciani, R. van Gelder, P. Geoerg, N. Hocaoglu, S. Laws, R. Lovreglio, Z. Miles, A. Nicolas, W. J. O'Toole, S. Schaap, T. Semmens, Z. Shahhoseini, R. Spaaij, A. Tatrai, J. Webster, and A. Wilson, “A roadmap for the future of crowd safety research and practice: Introducing the swiss cheese model of crowd safety and the imperative of a vision zero target,” *Safety Science*, vol. 168, p. 106292, 2023.

[18] J. Zhong, D. Li, W. Cai, W.-N. Chen, and Y. Shi, “Automatic crowd navigation path planning in public scenes through multiobjective differential evolution,” *IEEE Transactions on Computational Social Systems*, vol. 11, no. 1, pp. 905–918, 2024.

[19] H. Gu, S. Kim, and M. Jung, “Data-driven urban planning for proactive crowd management: Lessons from the 2022 seoul halloween crowd crush,” *Cities*, vol. 168, pp. 1–14, 2026.

[20] K. Rezaee, M. R. Khosravi, H. Attar, V. G. Menon, M. A. Khan, H. Issa, and L. Qi, “Iomt-assisted medical vehicle routing based on uav-borne human crowd sensing and deep learning in smart cities,” *IEEE Internet of Things Journal*, vol. 10, no. 21, pp. 18 529–18 536, 2023.

[21] Shanghai Municipal Public Security Bureau Huangpu Branch, “Notice on the issuance of the “Implementation Plan for Safety Management of Key Areas such as the Bund and Nanjing Road for the 2023 New Year”,” 2022, (in Chinese).

[22] CCTV, “Beijing dongcheng police take multiple measures to handle large crowds at nanluoguxiang,” Oct. 2021, (in Chinese).

[23] H. Yu, N. Jiang, M. Li, X. Jia, J. Shi, E. Wai Ming Lee, and L. Yang, “Regulating multi-directional passenger flow: Impact of obstacle position and flow level on pedestrian merging process,” *Tunnelling and Underground Space Technology*, vol. 157, p. 106336, 2025.

[24] D. Wu and H. Diao, “Optimization of security police deployment and scheduling for large-scale events,” *Xitong Gongcheng Lilun yu Shijian (Syst. Eng. — Theory & Pract.)*, vol. 42, no. 03, pp. 789–800, 2022, (in Chinese).

[25] N. Molyneaux and M. Bierlaire, “Controlling pedestrian flows with moving walkways,” *Transportation Research Part C: Emerging Technologies*, vol. 141, p. 103672, 2022.

[26] X. C. Liao, W. N. Chen, X. Q. Guo, J. Zhong, and D. J. Wang, “Drift: A dynamic crowd inflow control system using lstm-based deep reinforcement learning,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 55, no. 6, pp. 4202–4215, 2025.

[27] H. M. Al-Ahmadi, I. Reza, A. Jamal, W. S. Alhalabi, and K. J. Assi, “Preparedness for mass gatherings: A simulation-based framework for flow control and management using crowd monitoring data,” *Arabian Journal for Science and Engineering*, vol. 46, no. 5, pp. 4985–4997, 2021.

[28] L. Feng and E. Miller-Hooks, “A network optimization-based approach for crowd management in large public gatherings,” *Transportation Research Part C: Emerging Technologies*, vol. 42, pp. 182–199, 2014.

[29] R. X. Zhong, Y. P. Huang, C. Chen, W. H. K. Lam, D. B. Xu, and A. Sumalee, “Boundary conditions and behavior of the macroscopic fundamental diagram based network traffic dynamics: A control systems perspective,” *Transportation Research Part B: Methodological*, vol. 111, pp. 327–355, 2018.

[30] R. Zhao, Q. Liu, Q. Hu, D. Dong, C. Li, and Y. Ma, “Lyapunov-based crowd stability analysis for asymmetric pedestrian merging layout at t-shaped street junction,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 11, pp. 6833–6842, 2021.

[31] S. A. Wadoo and P. Kachroo, “Feedback control of crowd evacuation in one dimension,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 11, no. 1, pp. 182–193, 2010.

[32] W. Qin, “Unit sliding mode control for disturbed crowd dynamics system based on integral barrier lyapunov function,” *IEEE Access*, vol. 8, pp. 91 257–91 264, 2020.

[33] Y. Zhu, D. Zhao, and H. He, “Optimal feedback control of pedestrian flow in heterogeneous corridors,” *IEEE Transactions on Automation Science and Engineering*, vol. 18, no. 3, pp. 1097–1108, 2021.

[34] Y. Zhou, M. Zhong, Z. Li, X. Xu, F. Hua, and R. Pan, “Simulation-based adaptive optimization for passenger flow control measures at metro stations,” *Simulation Modelling Practice and Theory*, vol. 138, p. 103021, 2025.

[35] X. Yang, B. Shi, G. Zhang, and Y. Li, “Optimization modeling of automatic crowd regulation at bottlenecks of subway system: A model predictive control approach,” *Physics Letters A*, vol. 531, p. 130180, 2025.

[36] X. Yang, G. Zhang, B. Shi, C.-Z. Xie, and B. Zhang, “Partition independent control and collaborative optimization of high-density crowd in subway stations,” *Chaos, Solitons & Fractals*, vol. 200, p. 117108, 2025.

[37] M. A. Lopez-Carmona and A. Paricio-Garcia, “Crowd evacuation management with optimal time-multiplexed exit-choice recommendations through simulation optimization,” *Advanced Theory and Simulations*, vol. 7, no. 7, p. 2400265, 2024.

[38] M. A. Lopez-Carmona and A. Paricio Garcia, “Linear and nonlinear model predictive control (mpc) for regulating pedestrian flows with discrete speed instructions,” *Physica A: Statistical Mechanics and its Applications*, vol. 625, p. 128996, 2023.

[39] X. C. Liao, W. N. Chen, X. Q. Guo, J. Zhong, and X. M. Hu, “Crowd management through optimal layout of fences: An ant colony approach based on crowd simulation,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 9, pp. 9137–9149, 2023.

[40] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.

[41] K. Price, R. M. Storn, and J. A. Lampinen, *Differential Evolution: A Practical Approach to Global Optimization*. Springer Publishing Company, Incorporated, 2014.

[42] W. L. Liu, Y. Chen, X. Li, J. Zhong, R. Chen, and H. Jin, “Automatic optimization of guidance guardrail layout based on multi-objective evolutionary algorithm,” *Complex System Modeling and Simulation*, vol. 4, no. 4, pp. 353–367, 2024.

[43] L. F. Henderson, “The statistics of crowd fluids,” *Nature*, vol. 229, no. 5284, p. 381–383, 1971.

[44] R. L. Hughes, “A continuum theory for the flow of pedestrians,” *Transportation Research Part B: Methodological*, vol. 36, no. 6, pp. 507–535, 2002.

[45] L. Huang, S. C. Wong, M. Zhang, C.-W. Shu, and W. H. K. Lam, “Revisiting hughes' dynamic continuum model for pedestrian flow and the development of an efficient solution algorithm,” *Transportation Research Part B: Methodological*, vol. 43, no. 1, p. 127–141, 2009.

[46] R. M. COLOMBO, M. GARAVELLO, and M. LÉCUREUX-MERCIER, “A class of nonlocal models for pedestrian traffic,” *Mathematical Models and Methods in Applied Sciences*, vol. 22, no. 04, p. 1150023, 2012.

[47] B. Raimund, G. Paola, I. Daniel, and V. Luis Miguel, “A non-local pedestrian flow model accounting for anisotropic interactions and domain boundaries,” *Mathematical Biosciences and Engineering*, vol. 17, no. 5, p. 5883–5906, 2020.

[48] B. Maury and J. Venel, “A mathematical framework for a crowd motion model,” *Comptes Rendus Mathematique*, vol. 346, no. 23, p. 1245–1250, 2008.

[49] B. Maury, A. Roudneff-Chupin, F. Santambrogio, and J. Venel, “Handling congestion in crowd motion modeling,” *Networks and Heterogeneous Media*, vol. 6, no. 3, p. 485–519, 2011.

[50] R. Jordan, D. Kinderlehrer, and F. Otto, “The variational formulation of the fokker–planck equation,” *SIAM Journal on Mathematical Analysis*, vol. 29, no. 1, p. 1–17, 1998.

[51] A. Aw and M. Rascle, “Resurrection of ”second order” models of traffic flow,” *SIAM Journal on Applied Mathematics*, vol. 60, no. 3, p. 916–938, 2000.

[52] J.-P. Lebacque, S. Mammar, and H. Haj-Salem, “The aw–rascle and zhang's model: Vacuum problems, existence and regularity of the solutions of the riemann problem,” *Transportation Research Part B: Methodological*, vol. 41, no. 7, p. 710–721, 2007.

[53] E. Cristiani, F. S. Priuli, and A. Tosin, “Modeling rationality to control self-organization of crowds: An environmental approach,” *SIAM Journal on Applied Mathematics*, vol. 75, no. 2, pp. 605–629, 2015.

[54] F. Priuli, A. Tosin, and E. Cristiani, “On the control of self-organized systems: The case of pedestrian flows,” in *Proceedings of the 5th International Conference on Pedestrian and Evacuation Dynamics*, 2015.